KUKU AI — Microdrama Promo Intelligence Engine

Problem 1: Automated Screenplay Analysis, Hook/Cliffhanger Detection & Ad Funnel Stratification

Candidate Solution Doc
Scope: 50 Episodes ("I Married a Broken Soldier")

1. PROBLEM FRAMING, BUSINESS CONTEXT & SCOPING

Why Problem 1? Kuku's primary growth engine is short-form vertical microdramas. Top-of-funnel customer acquisition on TikTok, Instagram Reels, and YouTube Shorts relies entirely on **20–60 second promo ads**. Currently, human editors manually scrub through 50+ episodes per show every week—an expensive 4-to-6 hour bottleneck per series. Automating promo discovery directly accelerates content velocity, lowers CPA (cost per acquisition), and drives subscriber conversion.

Core Assumptions Made:

- **The 3-Second Hook Rule:** A scroller decides to stay or swipe within 3.0s. The opening must contain an immediate sensory trigger: physical action (slap, shove, drawn sword), high-arousal dialogue (accusation, betrayal, disownment), or shocking visual reveal.
- **Boundary Precision:** Promo cuts cannot slice mid-sentence or mid-breath. Cuts snap deterministically to natural dialogue pauses or scene boundaries.
- **Standalone Comprehension:** Promos must hook cold scrollers who have never seen the show. Intricate late-season political lore without clear emotional stakes is penalized for top-of-funnel ad spend.

2. SYSTEM ARCHITECTURE: BALANCING DETERMINISTIC CODE VS. AI

A naive LLM approach—stuffing 100 minutes of screenplay text into a prompt—is slow, expensive, and inaccurate at calculating exact durations. We engineered a **hybrid two-tier pipeline** separating mathematical constraints from creative semantic judgment:

⚙️ Tier 1: Deterministic Code & Heuristics (Python)

- **Regex AST Parser:** Parses scene headings, speaker emotions, and millisecond timestamps into typed dataclasses in **0.07s for 50 episodes**.
- **Sliding Candidate Generator:** Generates valid windows strictly bound between **20.0s and 60.0s** across block boundaries.
- **Acoustic & Action Heuristic Filter:** Scans the first 3.5s and final 4.5s for 60+ conflict triggers (slaps, gasps, drones, "Cut to black").
- **Non-Maximum Suppression (NMS):** Suppresses temporal overlaps (IoU > 0.50), eliminating duplicate cuts of the same scene.

🤖 Tier 2: Multi-Model LLM Reasoning (AI)

- **Hook Potency Scoring (0–10):** Evaluates visceral scroll-stopping power in the first 3 seconds.
- **Cliffhanger Tension Scoring (0–10):** Measures unresolved narrative tension compelling the viewer to install Kuku to see what happens next.
- **Standalone Coherence Scoring (0–10):** Validates whether the conflict is universally understandable without prior lore.
- **Conversion Pitch Generation:** Produces the 1-line rationale for marketing media buyers.

Model-Agnostic Design: The system features a pluggable backend. It runs instantly offline via high-fidelity local heuristics (zero API cost) and connects seamlessly via one parameter to **Google Gemini Flash**, **OpenAI GPT-4o-mini**, or custom enterprise endpoints during the live interview.

3. ACQUISITION FUNNEL: WHICH PROMOS TO POST FIRST & WHY

Not all promos serve the same purpose. Rather than a flat list, our ranker stratifies outputs into a **three-tier marketing funnel**:

Funnel Tier	Target Audience	Primary Characteristics	Recommended Posting Order & Strategy
Tier 1: High-Velocity Cold Ads	Cold Scrollers (TikTok, IG Reels, YT Shorts)	Extreme shock in 0–3s, universal archetypes (betrayal at altar, cruel in-laws, unmasking).	**Post 1st:** Drives mass top-of-funnel CTR. Lowest CPA; stops scrollers who know nothing about Kuku.
Tier 2: Escalation & Retargeting	Intrigued Viewers (Watched > 50% of Ad 1)	Unbearable cliffhanger cuts mid-sentence; life-or-death ultimatum or revenge declaration.	**Post 2nd:** Retargeting ads driving app install to unlock Episode 2 and beyond.
Tier 3: Organic Episodic Promos	Active Series Followers & Community	Nuanced romance, character chemistry, secondary plotlines.	**Post Later:** Organic posting on show accounts to sustain mid-season retention.

Key Strategic Insight: Top 3 overall promos identified across the 50 episodes: **#1 Ep 38 (00:45–01:07 | 21.5s)** — Juliana's desperate declaration: "*I will not give them my husband*" (Score: 8.0/10). **#2 Ep 1 (01:12–01:35 | 23.1s)** — The altar abandonment

confrontation: "*He was never going to marry you*" (Score: 7.2/10). **#3 Ep 36 (02:23–03:20 | 57.5s)** — The Duke's identity reveal: "*...their dead Duke is coming home*" (Score: 7.6/10).

KUKU AI — Quality Audits, Limits & Scalability

Failure Analysis, Human-in-the-Loop Workflow & Production Economics

4. HONEST QUALITY CHECKING: WHERE DOES THE TOOL PICK BADLY?

No algorithmic tool is infallible. We audited all 150 extracted promos and identified **4 systemic failure modes** where automated text-based analysis degrades:

Failure Mode	Root Cause in Screenplay Text	Real Example from Dataset	Human-in-the-Loop Mitigation
FM1: Late-Show Lore Bleed	High cliffhanger score in late episodes (Ep 35+) that relies on intricate relationship lore unknown to a cold scroller.	**Ep 46 (00:17–00:47):** High Priest decree regarding vagrant marriage legality. High drama, but cold viewers lack context.	Restrict Tier-1 ad spend to Episodes 1–10, or insert a 2-second on-screen context overlay card.
FM2: Vertical Pacing Lag	Clip duration is 40–60s with sparse dialogue, relying on silent atmospheric staring that works on TV but causes instant swipe-away on TikTok.	**Ep 21 (00:00–00:40):** Atmospheric barn setup with 3 dialogue lines. Pacing is too languid for vertical video.	Automated filter penalizes candidates with dialogue turn density < 0.15 turns/sec; editor trims silent pauses.
FM3: Subtle Opening Setup	Clip opens with polite or conversational setup before the explosive confrontation hits 5s in, missing the 3-second hook window.	**Ep 1 (00:11–00:43):** Juliana enters cathedral; dramatic reveal occurs at 00:18. Opening 3s is just church doors.	Snip opening 4 seconds; start video in medias res on the Matriarch's confrontation line.
FM4: Physical/Lexical Disconnect	High-stakes dialogue that was delivered with deadpan or humorous visual staging in production.	Sarcastic lines ("*You're a genius*") misclassified as serious dramatic tension.	Editor verification dashboard with 1-click reject/trim before pushing to Meta/TikTok Ad Manager.

Human-in-the-Loop Workflow: The tool is not designed to replace the video editor; it serves as an **intelligent editorial co-pilot**. Instead of watching 100 minutes of footage, the editor opens the dashboard, reviews the top 3 pre-trimmed candidates per episode with pre-filled timestamps and rationale, and approves/refines them in under 5 minutes.

5. PRODUCTION ECONOMICS & SCALING TO EVERY SHOW KUKU MAKES

Kuku releases multiple new shows weekly. Our two-tier architecture guarantees near-zero latency and negligible compute costs:

⚡ Computational Throughput

- **50 Episodes Parsed & Scored: 2.28 seconds** on a standard consumer laptop.
- **Scaling to 100-Episode Season:** ~4.5 seconds execution time.
- **Zero GPU Requirement:** Pure CPU Python parsing; parallelizable across batch cloud workers.

💰 Unit Economics (Gemini 1.5 Flash / GPT-4o-mini)

- **Candidate Filtering:** NMS reduces raw token load by **95%** (evaluates ~150 candidates vs 3,700 blocks).
- **Tokens per Series:** ~45,000 input/output tokens.
- **Total API Cost per Series: $0.0034 (< 0.5 cents)**. 100 shows/yr = **$0.34 total**.

6. SUBMISSION DELIVERABLES & LIVE INTERVIEW VERIFICATION

All required deliverables have been implemented, verified, and packaged for immediate evaluation:

- **Working Interactive Prototype:** Full-featured Streamlit application supporting episode browsing, real-time parameter tuning (duration, hook windows), screenplay upload for new shows, and JSON/CSV data export.
- **Unedited Raw Outputs:** Complete 50-episode extraction datasets generated and formatted in `outputs/raw_promos_all_episodes.json` and `outputs/raw_promos_all_episodes.csv`.
- **Shortlist Interview Ready:** Built with modular interfaces. In the live interview, drop any unseen `.txt` screenplay or connect Kuku's proprietary LLM API key, and the pipeline runs end-to-end in seconds.